

NUTRITION ASSESSMENT QUESTIONNAIRE

Healthy Eating Index, Double Burden of Malnutrition and BMI among Adults

Respondent ID	_____	Date (DD/MM/YY)	____ / ____ / ____
District / area	_____	Interviewer code	_____

Enumerator: ask every item; circle/tick one coded response unless instructed otherwise. Added variables are marked with +.

A SOCIODEMOGRAPHIC

D HEI - FATS, SODIUM & ADDED SUGARS

A1. Name: _____

A2. Age (completed years): _____ (eligible: adult >=18 years)

A3. Sex: ☐ Male [0] ☐ Female [1] ☐ Other [2]

A4. Religion: ☐ Islam [0] ☐ Hinduism [1] ☐ Christian [2] ☐ Buddhism [3] ☐ Other [4]

A5. Profession / occupation: ☐ Service [0] ☐ Business [1] ☐ Agriculture [2] ☐ Labour [3] ☐ Homemaker [4] ☐ Student [5] ☐ Unemployed [6] ☐ Other [7]: _____

A6. Personal monthly income (BDT): _____ ☐ None [0] ☐ <=10k [1] ☐ 10,001-20k [2] ☐ 20,001-40k [3] ☐ >40k [4]

A7. Marital status: ☐ Married [0] ☐ Unmarried [1] ☐ Divorced/separated [2] ☐ Widowed [3]

A8. Educational background: ☐ None [0] ☐ Primary [1] ☐ SSC [2] ☐ HSC [3] ☐ Graduate [4] ☐ Postgraduate [5]

A9. Family size: _____ persons

A10. Total monthly family income (BDT): _____ ☐ <=10k [0] ☐ 10,001-20k [1] ☐ 20,001-40k [2] ☐ >40k [3]

A11. Family type: ☐ Nuclear [0] ☐ Joint/extended [1]

A12. Food insecurity (not enough food) in past 4 weeks: ☐ Absent [0] ☐ Present [1]

A13.+ Residence: ☐ Rural [0] ☐ Urban [1] ☐ Semi-urban [2] ☐ Slum [3]

A14.+ Smoking status: ☐ Never [0] ☐ Former [1] ☐ Current [2]

A15.+ If biologically applicable, currently: ☐ Neither [0] ☐ Pregnant [1] ☐ Lactating [2] ☐ N/A [9]

B ANTHROPOMETRY & BMI

Measure without shoes; record to one decimal where possible.

B1 Height	_____ cm	B2 Weight	_____ kg
B3 BMI	_____ kg/m²	B4 Energy requirement	_____ kcal/day
B5 Energy intake	_____ kcal/day	B6+ Waist circumference	_____ cm
B7+ Mid-upper arm circumference	_____ cm	B8+ Blood pressure	____/____ mmHg

BMI class (researcher): ☐ <18.5 Underweight [0] ☐ 18.5-24.9 Normal [1] ☐ 25.0-29.9 Overweight [2] ☐ >=30 Obesity [3]

C HEI - FOOD-GROUP CONSUMPTION

F1 frequency: [0] Never; [1] 1-2 d/wk; [2] 3-4 d/wk; [3] 5-6 d/wk; [4] Daily; [9] DK. Write 0-7 where 'days' is requested.

C1. Unintentional weight loss in past 3-6 months? ☐ No [0] ☐ Yes [1]; if yes: _____ kg

C2. Days in past 7 days ate whole fruits (fresh/frozen/canned in juice; not juice): ____/7

C3. Servings of 100% fruit juice typically consumed per day: _____ servings/day

C4. Days in past 7 days ate non-starchy vegetables (broccoli, tomato, carrot, cabbage): ____/7

C5. Dark green leafy vegetables (spinach, kale, amaranth, mustard greens)? F1: _____

C6. Cooked dried beans, lentils or peas? F1: _____

C7. Whole grains (brown rice, whole-wheat bread, oats, millet, sorghum)? F1: _____

C8. Refined grains (white rice/bread, noodles, plain biscuits)? F1: _____

C9. Dairy (milk, yoghurt, cheese, calcium-fortified plant milk)? F1: _____

C10. Lean protein (chicken, fish, eggs, tofu, low-fat meat)? F1: _____

C11. Processed/preserved meat (sausage, salami, smoked fish, canned meat)? F1: _____

Use F1 above unless another response is shown.

D1. Primary household cooking fat/oil: ☐ Soy [0] ☐ Mustard [1] ☐ Palm [2] ☐ Rice bran [3] ☐ Ghee/butter [4] ☐ Other [5]: _____

D2. Deep-fried foods (fried snacks/fish, pakora, chips)? F1: _____

D3. Fast food or commercially processed meals? F1: _____

D4. Adds extra salt at table or during cooking? F1: _____

D5. Sugar-sweetened beverages (soda, fruit drink, sweet tea/coffee)? F1: _____

D6. Sweets, cakes, pastries or confectionery? F1: _____

D7. Nuts, seeds or nut butters? F1: _____

D8. Fatty fish (hilsa, mackerel, sardine, salmon)? F1: _____

D9. Reads nutrition/ingredient labels before purchase? ☐ Never [0] ☐ Sometimes [1] ☐ Usually/always [2] ☐ N/A [9]

E FANTA/FAO DIETARY DIVERSITY - 24-HOUR RECALL

Ask for all foods/drinks eaten yesterday; probe mixed-dish ingredients. Mark Yes [1] or No [0]. The 16 standard prompts aggregate to the 12-group HDDS below.

Food group	Bangladesh-relevant examples	No/Yes
E1 Cereals	rice, wheat/atta, maize, bread, noodles	☐ 0 ☐ 1
E2 White roots & tubers	potato, white yam/cassava, taro	☐ 0 ☐ 1
E3 Vitamin A-rich veg/tubers	pumpkin, carrot, orange sweet potato	☐ 0 ☐ 1
E4 Dark green leafy veg	spinach, amaranth, mustard/moringa leaves	☐ 0 ☐ 1
E5 Other vegetables	tomato, onion, eggplant, cabbage	☐ 0 ☐ 1
E6 Vitamin A-rich fruits	ripe mango/papaya, cantaloupe, apricot	☐ 0 ☐ 1
E7 Other fruits	banana, apple, guava, citrus, watermelon	☐ 0 ☐ 1
E8 Organ meat	liver, kidney, heart, other offal	☐ 0 ☐ 1
E9 Flesh meat	beef, mutton/goat, chicken, duck, pork	☐ 0 ☐ 1
E10 Eggs	chicken, duck, quail or other eggs	☐ 0 ☐ 1
E11 Fish & seafood	fresh/dried fish, shrimp, crab/shellfish	☐ 0 ☐ 1
E12 Legumes, nuts & seeds	dal, lentils, beans, chickpeas, peanuts, seeds	☐ 0 ☐ 1
E13 Milk & milk products	milk, yoghurt, cheese; exclude milk in tea	☐ 0 ☐ 1
E14 Oils & fats	oil, ghee, butter added/used in cooking	☐ 0 ☐ 1
E15 Sweets	sugar/honey, sweets, cake, sweet drinks	☐ 0 ☐ 1
E16 Spices/condiments/beverages	spices, sauces, tea/coffee, alcohol	☐ 0 ☐ 1

HDDS aggregation (12): **E1; E2; E3-E5; E6-E7; E8-E9; E10; E11; E12; E13; E14; E15; E16. Total HDDS: _____/12**

Source: FANTA HDDS Indicator Guide v2 (2006), as reproduced/adapted in FAO dietary-diversity guidance. No universal low/medium/high HDDS cut-offs are established; analyze the score continuously or by sample distribution. This is dietary diversity, not HEI scoring.

F MEAL PATTERNS & EATING BEHAVIOURS

F2: [0] Never; [1] 1-2 d/wk; [2] 3-4 d/wk; [3] 5-6 d/wk; [4] Daily.
F1. Main meals typically consumed per day: ☐ 1 [1] ☐ 2 [2] ☐ 3 [3] ☐ >=4 [4]
F2. Regularly eats breakfast within 2 hours of waking? ☐ No [0] ☐ Yes [1]
F3. Days/week skips at least one main meal: ____ /7
F4. Home-cooked meals vs outside food: ☐ Mostly home [0] ☐ Equal [1] ☐ Mostly outside [2]
F5. Access to clean, safe drinking water at home? ☐ No [0] ☐ Yes [1]
F6. Plain water consumed daily: ____ cups/day
F7. Prescribed or self-selected dietary pattern? ☐ No [0] ☐ Yes [1]: ☐ Vegetarian ☐ Low-salt ☐ Diabetic ☐ Weight-loss ☐ Other: ____
F8. Late-night meals/snacks after 9 PM? F2: ____
F9. Snacks between main meals? F2: ____

G DOUBLE BURDEN OF MALNUTRITION

For G1-G8: ☐ No [0] ☐ Yes [1] ☐ Don't know [9], unless otherwise stated.
G1. Previously diagnosed with undernutrition, protein-energy malnutrition (PEM) or wasting? Code: ____
G2. Ever diagnosed with micronutrient-deficiency anaemia (iron, folate or B12)? Code: ____
G3. Significant fatigue, weakness or poor wound healing recently? Code: ____
G4. Current/history of food insecurity (not enough food to eat)? Code: ____
G5. Adequate dietary protein (assessor estimate from recall)? ☐ Inadequate [0] ☐ Adequate [1] ☐ Unclear [9]
G6. Clinically diagnosed Type 2 diabetes mellitus? Code: ____
G7. Diagnosed hypertension (BP >=140/90 mmHg)? Code: ____
G8. Current/prior cardiovascular disease (heart attack, stroke, heart failure)? Code: ____
G9. Feels they consume more calories than needed? ☐ Never [0] ☐ Sometimes [1] ☐ Often [2] ☐ DK [9]
G10.+ Currently takes medicine for diabetes, hypertension or lipids? ☐ No [0] ☐ Yes [1]: ____
G11.+ Other diagnosed chronic condition affecting diet/weight? ☐ No [0] ☐ Yes [1]: ____

H PHYSICAL ACTIVITY & SEDENTARY BEHAVIOUR

Record days (0-7); a valid activity bout should last at least 10 minutes.
H1. Days/week of moderate-intensity activity (brisk walking, cycling, household work): ____ /7; average ____ min/day
H2. Days/week of vigorous-intensity activity (running, heavy lifting, fast cycling): ____ /7; average ____ min/day
H3.+ Typical sitting/reclining time while awake: ____ hours/day

FIELD ENUMERATOR CHECKS

X1.+ Interview language: ☐ Bangla [0] ☐ English [1] ☐ Other [2]: ____
X2.+ Recall day unusual (feast, fasting, illness, travel)? ☐ No [0] ☐ Yes [1]: ____
X3.+ Form complete and reviewed? ☐ No [0] ☐ Yes [1]

I PSYCHOSOCIAL & MENTAL-HEALTH FACTORS

I1-I2 (past 2 weeks): [0] Not at all; [1] Several days; [2] >half the days; [3] Nearly every day.
I1. Felt down, depressed or hopeless? Code: ____
I2. Felt nervous, anxious or on edge? Code: ____
I3. Overall stress in past month: ☐ None [0] ☐ Low [1] ☐ Moderate [2] ☐ High [3] ☐ Very high [4]
I4. Eats more than planned when stressed/emotionally upset? ☐ No [0] ☐ Sometimes [1] ☐ Yes [2]
I5. Positive relationship with food (without guilt, shame or fear)? ☐ No [0] ☐ Sometimes [1] ☐ Yes [2]
I6. Sleep per night: ____ hours

J NUTRITION KNOWLEDGE, ATTITUDES & SELF-EFFICACY

For J1-J5: [1] Strongly disagree; [2] Disagree; [3] Neutral; [4] Agree; [5] Strongly agree.
J1. I understand what a balanced diet means and which food groups should be included. Circle: 1 2 3 4 5
J2. I know the recommended daily servings of fruits and vegetables for an adult. Circle: 1 2 3 4 5
J3. I can identify foods high in saturated fat, added sugar or sodium on a food label. Circle: 1 2 3 4 5
J4. I understand the link between poor diet and chronic diseases such as diabetes and heart disease. Circle: 1 2 3 4 5
J5. Good nutrition is a priority for me compared with other aspects of my daily life. Circle: 1 2 3 4 5

CONSENT VOLUNTARY PARTICIPATION

I have been informed about the purpose of this study. I voluntarily agree to participate and understand that my information will be kept confidential and used only for research purposes.
C0. Consent given: ☐ No [0] - STOP ☐ Yes [1]
Respondent signature/thumbprint: _____ Date: _____
Interviewer name/signature: _____ Date: _____

NOTES ENUMERATOR / ASSESSOR COMMENTS

Data-entry note: use the bracketed numeric codes; blank = missing, and [9] = don't know/not applicable where supplied. HEI-related questions require a prespecified scoring protocol; the FANTA/FAO HDDS must not be reported as an HEI score.

OFFICE SCORING & DATA-ENTRY CONTROL

Respondent ID		BMI	
BMI class code		HDDS (0-12)	
HEI score*	_____/100	Double-burden flag	☐ No [0] ☐ Yes [1]
Entered by/date		Verified by/date	

*Complete only using the study's approved HEI scoring protocol and quantitative dietary data.